# HOW THE AMR KNOWS WHERE IT IS

MIND MAP · EVERY BLOCK IN THE SLAM STACK AND WHO OWNS IT

## FRONT END · RUNS IN EVERY MODE

HARDWARE

### SENSORS

RAW MEASUREMENTS

| | |
|---|---|
| NANOSCAN3 | 2-D LiDAR, 190° sector, ~17 Hz → `/scan` |
| ENCODERS ×2 | BLV-R motor position over CANopen → `/wheel_states` |
| MLS GYRO | Yaw rate, CAN node 10 → `/imu/data_raw` |
| GLS621 | Floor-label camera, ~14.6 Hz → `/gls/reading` |

FEEDS

OWN CODE

### PRE-PROCESSING

AMR_BASE · AMR_LOCALIZATION

| | |
|---|---|
| SCAN GATE | TF-ready scans only, thinned to ~8.5 Hz |
| WHEEL ODOM | Midpoint integration of encoder deltas → x, y, θ |
| GYRO BIAS | 2 s still-window average subtracted (raw +0.057 °/s) |
| LABEL READ | Codes of one image → label ID + camera pose |

FEEDS

ROS PKG

### ODOMETRY FUSION

ROBOT_LOCALIZATION EKF

| | |
|---|---|
| FUSES | Forward speed $v_x$ + yaw rate only; never absolute yaw |
| RATE | 50 Hz, 2-D mode |
| OUTPUT | `odom → base_footprint`: smooth, but drifts |

## SLAM ON GVIEVO-01

2-D LIDAR GRAPH SLAM + PARTICLE FILTER

Build the map once in a supervised survey, then localize on the saved, approved map on every run. Floor labels check the pose where the vehicle stops.

SURVEY → SLAM_TOOLBOX | RUN → AMCL

## BACK END · SURVEY OR PRODUCTION, NEVER BOTH

ROS PKG · SURVEY

### MAPPING

SLAM_TOOLBOX · ONLINE ASYNC

| | |
|---|---|
| FRONT END | Correlative scan matching against recent scans |
| BACK END | Pose graph + loop closure, optimised by Ceres (LM) |
| OUTPUT | `/map` 5 cm grid · `map → odom` 20 Hz |

SAVE

OWN CODE

### MAP REVISIONS

MAPPING_SESSION_NODE · MAP_BUNDLE

| | |
|---|---|
| SURVEY | IDLE → MAPPING → RETURN_REVIEW → SAVING → SAVED |
| SAVED | map.pgm/.yaml + pose graph + hashed manifest |
| RULE | Atomic rename to rev N; never rewritten |
| LABELS | Label set per map rev: HOME + trolley stops |

LOAD

ROS PKG + OWN · RUN

### LOCALIZATION

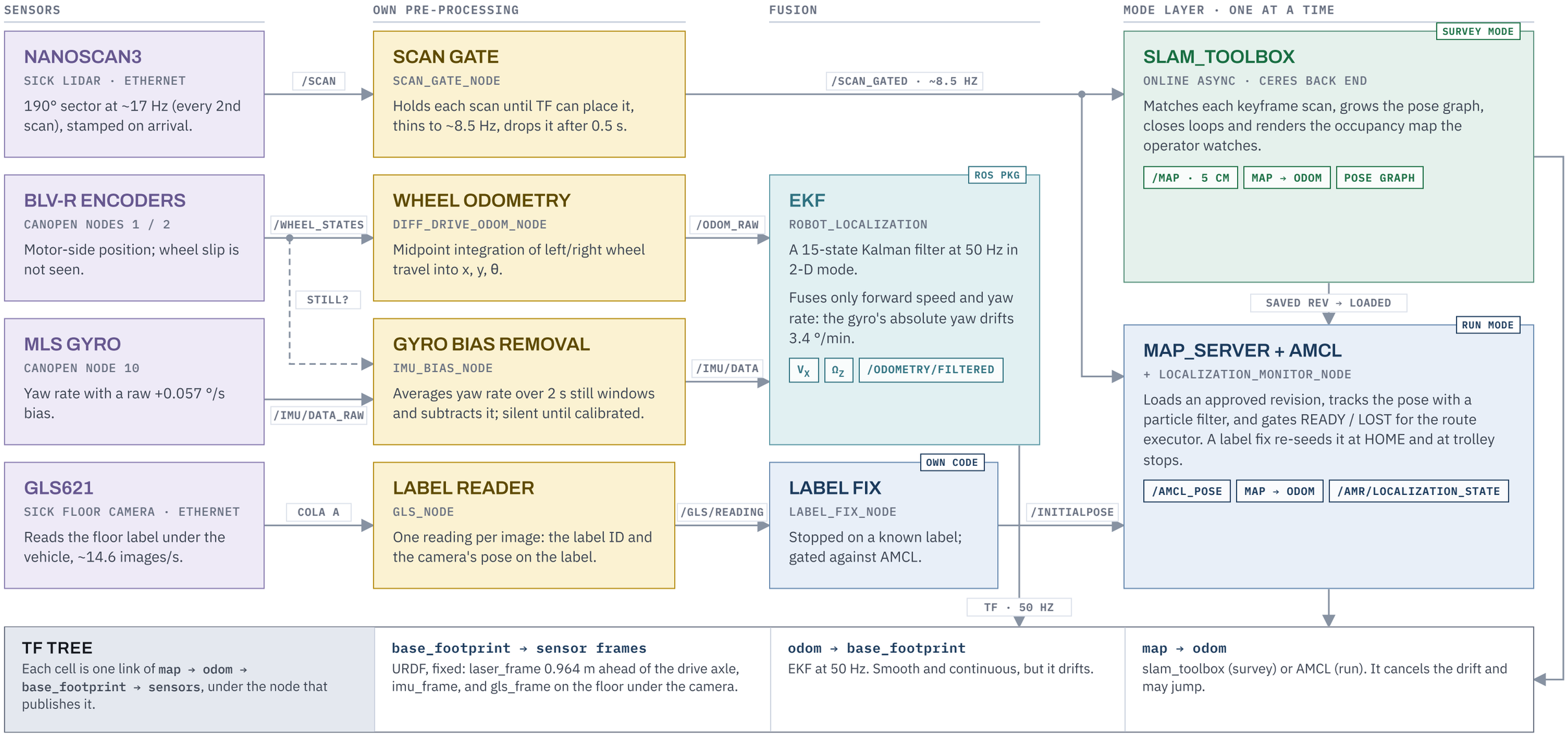
AMCL + LOCALIZATION MONITOR

| | |
|---|---|
| FILTER | 500–2 000 particles, likelihood field |
| GATE | READY needs settled checks + operator OK or a HOME label fix |
| LABELS | Re-zeros at HOME and at trolley stops |
| LOSS | No random recovery: LOST stops the robot |

SOURCE: AMR_WS/SRC · SLAM_MAPPING.YAML · AMCL.YAML · EKF.YAML · PROFILES/AGV-01.JSON GLS.*

# SENSOR-TO-POSE DATA FLOW

ROS 2 NODES · TOPICS · THE TF TREE THEY BUILD




CONSUMERS OF /SCAN_GATED: SLAM_TOOLBOX, AMCL, COSTMAP, MONITOR · WEB LIVE VIEW STAYS ON RAW /SCAN

# INSIDE SLAM_TOOLBOX

GRAPH SLAM LOOP · ONE PASS PER KEYFRAME SCAN · SLAM_MAPPING.YAML

FRONT END · WHERE AM I RELATIVE TO THE LAST SCANS?

BACK END · KEEP THE WHOLE GRAPH CONSISTENT

STEP 1

## KEYFRAME GATE

A gated scan (~8.5 Hz) becomes a keyframe only after 0.2 m or 0.2 rad of travel, and at most every 0.2 s.

STEP 2

## ODOMETRY PRIOR

Reads **odom → base_footprint** at the scan's timestamp: the EKF's guess of how far the robot moved.

STEP 3

## SCAN MATCHING

Slides the scan over a map of the last 10 scans: ±0.25 m at 1 cm, ±20° at 2°, then 0.2° fine.

STEP 4

## GROW THE GRAPH

Adds the matched pose as a node, plus an edge to the previous node carrying relative motion and its covariance.

NEW NODE

STEP 5

## LOOP CLOSURE

Searches old nodes within 3 m. A chain of ≥ 10 scans must match: coarse response ≥ 0.35, then fine ≥ 0.45.

STEP 6

## OPTIMISE

Ceres moves every node at once to minimise total edge error. Levenberg–Marquardt with a sparse Cholesky solve.

STEP 7

## PUBLISH

Re-renders the 5 cm occupancy grid at 1 Hz and publishes **map → odom** at 20 Hz to cancel odometry drift.

NEXT KEYFRAME

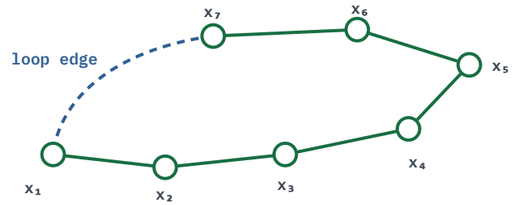
NO LOOP FOUND · SKIP STEP 6

## SURVEY LIFECYCLE

MAPPING_SESSION_NODE WRAPS THE LOOP ABOVE

| IDLE | MAPPING | RETURN_REVIEW | SAVING | SAVED |
|---|---|---|---|---|
| Readiness checks; start mark, and the HOME label if one is seen | Operator drives the survey; loop runs | Back at the mark: closure dx, dy, dθ shown | Map lock, pause SLAM, serialise graph, verify hashes | Atomic rename to rev N; HOME → label set rev 1 |

## THE MATH




### POSE-GRAPH OPTIMISATION

$$x^* = \underset{x}{\operatorname{argmin}} \sum_{(i,j)} e_{ij}^{\top}\,\Omega_{ij}\,e_{ij}$$

$$e_{ij} = z_{ij} \ominus (x_i^{-1} \oplus x_j)$$

x are the poses (x, y, θ), z is the relative pose measured by scan matching, and Ω is its inverse covariance. A confident edge pulls harder.

### EACH LEVENBERG–MARQUARDT STEP

$$(J^{\top}\Omega J + \lambda I)\,\Delta x = -J^{\top}\Omega e$$

A large λ takes small, cautious steps and a small λ approaches Gauss–Newton. $J^{\top}\Omega J$ is sparse, so sparse Cholesky solves it. There is no robust loss, so one false loop edge counts in full.

### SCAN-MATCH RESPONSE

$$\text{score}(p) = \text{penalty}(p) \cdot 1/N \sum_k \tilde{M}(T_p \cdot b_k)$$

$b_k$ are the beam endpoints and $\tilde{M}$ is the occupancy of recent scans smeared by 0.1 m. The penalty shrinks the score the further p strays from the odometry prior (distance 0.5, angle 1.0).

SEARCH WINDOWS: CORRELATION 0.5 M @ 0.01 M · LOOP 8 M @ 0.05 M · ANGLES 0.349 / 0.0349 / 0.00349 RAD

# LOCALIZING ON A SAVED MAP

AMCL PARTICLE FILTER · READINESS GATE · PRODUCTION RUNS

NAV2_AMCL · ONE CYCLE PER 5 CM OR 0.03 RAD OF TRAVEL

PREDICT

## MOVE THE PARTICLES

DIFFERENTIAL MOTION MODEL

Shifts every particle by the EKF odometry increment, plus noise scaled by $\alpha_1$–$\alpha_4$ = 0.1.

NEW CLOUD

FOR EACH PARTICLE

## PARTICLE CLOUD

500 – 2 000 POSE HYPOTHESES



/AMCL_POSE + COV · MAP→ODOM

RESAMPLE

## KEEP THE LIKELY ONES

KLD-ADAPTIVE · EVERY 2 UPDATES

Draws a new set in proportion to weight. The count adapts between 500 and 2 000 (error 0.05, 99 % confidence).

The best cluster gives the pose and its covariance.

WEIGHTS

WEIGHT · LIKELIHOOD_FIELD_PROB

## SCORE AGAINST THE MAP

Scores 180 beams by distance to the nearest mapped wall; beam skipping ignores unmapped clutter.

$$p(z_k) = z_{hit} \cdot N(d_k; 0, 0.1^2) + z_{rand} / z_{max}$$

OWN CODE

## LOCALIZATION MONITOR

LOCALIZATION_MONITOR_NODE · READINESS.PY

UNLOCALIZED → /INITIALPOSE → CHECKING → SETTLED 2 S + OPERATOR OK / LABEL FIX → READY → ANY TRIGGER → LOST

READY needs every automatic check to hold for 2 s, plus a confirm: the operator's (live scans line up with the map) or a HOME label fix that still agrees with AMCL. While CHECKING, AMCL is asked for an in-place update every 0.5 s, so a parked vehicle converges without a nudge. A new initial pose restarts CHECKING.

**COVARIANCE**
xx, yy ≤ 0.05 m² · yaw ≤ 0.03 rad². LOST if exceeded for 1 s.

**POSE JUMP**
`map→odom` moves > 0.15 m or > 5°. Ignored for 1.5 s after an initial pose.

**LONG BEAMS · LOSS TRIGGER**
`r_meas > r_map + tol` and the endpoint lands in mapped free space. LOST if > 15 % for 1 s.

**SCAN MATCH**
≥ 60 % of endpoints near mapped obstacles. Gates either confirm, operator or label.

**FRESHNESS**
scan 0.15 s · wheels 0.10 s · IMU 0.20 s (+0.6 s while drives power up) · TF 0.20 s · AMCL 5 s.

**DYNAMIC AREAS**
Beams in marked dynamic zones are left out of both scan fractions.

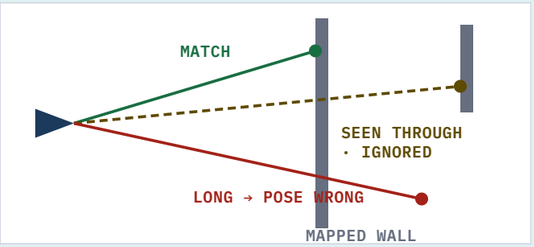
OWN CODE

## SCAN VS MAP

SCAN_CONSISTENCY.PY

Ray-casts the map from the AMCL pose to get each beam's expected range, then compares it with the measured range.

MATCH ≥ 60 % · LONG ≤ 15 %




**NO RANDOM RECOVERY** recovery_alpha_slow = recovery_alpha_fast = 0: AMCL never re-scatters particles, and a LOST robot stops until the operator gives a new initial pose. **LOST MEANS STOP, NOT GUESS**

SOURCE: AMCL.YAML · LOCALIZATION/READINESS.PY · SCAN_CONSISTENCY.PY · GLS-WAIT-CODING-PLAN §7 E

# FLOOR-LABEL FIXES

GLS621 CAMERA · LABEL SET PER MAP · A POSE CHECK WHERE THE VEHICLE STOPS

## CAMERA · READ-ONLY, OUTSIDE THE SAFETY PATH

HARDWARE

### GLS621

SICK FLOOR CAMERA · 192.168.4.1:2112

Looks down from ~203 mm and decodes the Data Matrix codes of one multi-code label: ~14.6 images/s, 4–5 codes each. Sends nothing when no label is in view.

TELEGRAMS

OWN CODE

### GLS_NODE

GLSLINK THREAD · NEVER WRITES THE CAMERA

| | |
|---|---|
| **VERIFY** | Settings read back at every connect; frames dropped until they match |
| **IMAGE** | Codes sharing one $FAMS → median x, y, angle; two labels = invalid |
| **OUTPUT** | Label ID + $T_{LC}$; invalid at 5 Hz after 0.5 s of silence |

OWN CODE

### LABEL SET

<MAP>/LABELS/REVN.YAML · PER MAP REVISION

| | |
|---|---|
| **HOME** | x0000y0000 · optional · read at survey start |
| **STOPS** | UBC x0100 · ITG x0200 · PISTON x0300 |
| **RECORD** | "Record label here": parked, READY, 15 frames → $T_{ML}$ from AMCL |

/GLS/READING

T_ML

## LABEL_FIX_NODE · ONLY WHILE STATIONARY

OWN CODE

### ACCEPT A READING

LABEL_FIX.PY · PURE, CLOCK-FED

| | |
|---|---|
| **STILL** | Wheels at rest for 1 s |
| **CODES** | ≥ 2 per image; 15 frames within 2 mm / 0.5° |
| **LABEL** | In the mission's label set, and the one expected at this stop |
| **GATE** | \|AMCL – GLS\| ≤ 0.30 m / 10°; at boot on HOME no prior is needed, the ID is the evidence |
| **DEADBAND** | A residual of a few mm: logged, not snapped |

Every decision goes to **/amr/label_fix** and the drift log: the residual at each stop is the drift measurement.

APPLY

APPLY

### SNAP AND CONFIRM

1 · Publish **/initialpose** at $T_{MB}$, σ ≥ 5 mm / 0.5°.

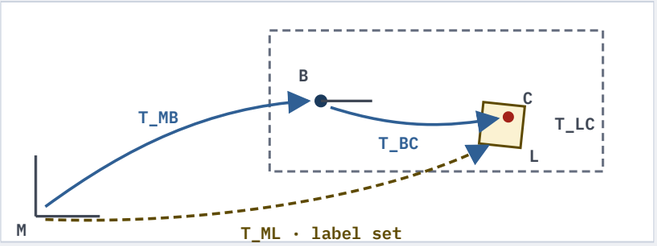
2 · AMCL re-seeds its particles there; the monitor restarts CHECKING.

3 · A fix that still agrees confirms READY without the operator (LOC_CONFIRMED_BY_LABEL). Scan match ≥ 60 % still applies.

GLS.CORRECT_AT_HOME | GLS.CORRECT_AT_WAITS

## FRAMES · SE(2) ON THE FLOOR

### WHERE THE POSE COMES FROM




```
record  T_ML = T_MB ⊕ T_BC ⊕ T_LC⁻¹
fix     T_MB = T_ML ⊕ T_LC ⊕ T_BC⁻¹
```

T_BC is the camera mount (**gls.mount_***), T_LC one reading, T_ML the label's entry in the set. A label ID is a name: no position is read from its digits.

### WHEN A FIX APPLIES

| WHERE | LABEL | AMCL PRIOR | THEN |
|---|---|---|---|
| Boot on HOME | HOME | none needed | **snap + confirm** |
| Parked at HOME after a run | HOME | ≤ 0.30 m / 10° | **re-zero** |
| WAITING at a trolley stop | its own | ≤ 0.30 m / 10° | **snap, then Continue** |
| Any other time | — | — | **log only** |

**DISAGREE MEANS STOP** GLS and AMCL over 0.30 m / 10° apart: LABEL_DISAGREE, Continue refused, no snap. A label not seen only warns (LABEL_NOT_SEEN). **LABELS CHECK, NEVER STEER**

SOURCE: GLS-WAIT-CODING-PLAN.MD §6–7 · GLS6-SYSTEM.MD §3 · PROFILES/AGV-01.JSON GLS.* · NO FIX WHILE MOVING, NONE INTO SLAM_TOOLBOX